Data Science and Machine Learning

Assignment-2

10/10

bitted to,
f. Anjana

Submitted by
Shyam kumar
S_3 MCA
Roll No-30

of Announcement: 7/11/22
of Submission: 11/11/22

2) Explain classification rule with example

classification rules represent knowledge in the form of logical if else statements that assign a class to unlabeled examples

They are specified in terms of an antecedent & consequent these form a hypothesis stating that "if this happens, then that happens"

The antecedent comprises certain combinations of feature values while the consequent specifies the class value to assign if the rules conditions are met

Example

if aerial = "Yes" ——→ Antecedent
    return class = "bird" —> consequent
else
    return class = "mammal"
end if

classification rule learning algorithm
a heuristic known as separate & conquer
The process involves identifying a
covers a subset of examples in the train
& then separating this partition from th

One Rule Algorithm (IR Algorithm)

- The way this algorithm works is si
for each feature, IR divides the data into
groups based on similar values of the feat
Then, for each segment the algorithm
predicts the majority class.
The error rate for the rule based
each feature is calculated & the rule wi
fewest errors is chosen as the one rule.

How do you pro

A decision tree can continue indefinitely choosing splitting features & dividing into smaller & smaller partitions until each example is perfectly classified or the algorithm runs out of features to split on

However if the tree grows overly large, many of the decisions it makes will be overly specific & the model will have been overfitted to the training data.

The process of pruning a decision tree involves reducing its size such that it generalizes better to unseen data.

One solution to this problem is to stop the tree from growing once it reaches a certain number of decisions or if the decision nodes contain only a small number of examples. This is called early stopping or pre-pruning the tree.

An alternative called post pruning is growing a tree that is too large, then using pruning criteria based on the error rates at nodes to reduce the size of the tree to an appropriate level.

This is often a more effective than pre-pruning because it is quite difficult to determine the optimal depth of a